



SigmaStar Display

Spinand 母片制作



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REVISION HISTORY

Revision No.	Description	Date
0.1	Initial release	07/26/2019
0.2	Beta release	01/14/2020



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1. SPINAND 母片制作方法

1.1. 基本文件主要有下面五个：

1.可执行程序 SstarMakeBin

2.初始化文件 SPINAND.INI(spinand)

SPINAND.INI 中的 nandid 需要根据实际 flash 的 datasheet 来填写, image_directory 为对应需要打包的 image 路径：

```
1  [path]
2  root_directory=./
3  image_directory=images/
4  script_file=auto_update_spinand.txt
5  outpath=./
6
7  [nand]
8  nandtype=SPINAND
9  nandid=0xC2-0x12
10
11 [env]
12 env_type=NANDRAW
13 env_part=ENV
14 #env_volume=MPOOL
15 #env_offset=0
16
17 [cis]
18 ecctype=10
19 nni=SPINANDINFO.sni
20 pni=PARTINFO.pni
21 #ppm=PAIRPAGEMAP_v2.ppm
22
23 [fcie]
24 type=fcie5
25
26 [boot]
27 #boottype=BFN
28 #miu=bfm_miu.bin
```

3.烧录器的配置文件 leap.def & snchk.def(leap 烧录器用 leap.def,其他烧录器一般用 snchk.def)

4.用来制作母片的 image 文件以及对应的烧录脚本，统一放在 images 文件夹下：

images	2020/3/18 11:05	文件夹	
auto_update_spinand.txt	2020/3/4 14:36	文本文档	1 KB
readme.txt	2020/3/4 15:20	文本文档	4 KB
SPINAND.INI	2020/3/18 11:08	配置设置	1 KB
SstarMakeBin	2020/3/4 12:43	文件	2,049 KB
leap.def	2020/3/17 20:53	DEF 文件	1 KB
snchk.def	2020/3/17 20:53	DEF 文件	1 KB

images 文件夹是从 project\image\output\images 下面的 images 拷贝过来的

5.配置分区和 image 路径的脚本 auto_update_spinand.txt

需要注意的是 auto_update_spinand.txt 下面 dynpart 这行的分区信息需要跟 uboot 下面的 mtdparts

保持一致：

```
SigmaStar #
SigmaStar # print
baudrate=115200
bootargs=console=ttyS0,115200 ubi.mtd=UBI,2048 root=ubi:rootfs rw rootfstype=ubifs init=/linuxrc rootwait=1 LX_MEM=0x7f00000 mma_heap=mma_heap_name0,miu=0,sz=0x3800000 mma_memblock
nd0:384k@1280k(IPL0),384k(IPL1),384k(IPL_CUST0),384k(IPL_CUST1),768k(UBOOT0),768k(UBOOT1),384k(ENV0),0x60000(Logo),0x500000(KERNEL),0x500000(RECOVERY),-(UBI)
bootcmd=dcache on; mw 1f001cc0 11; gpio out 8 0; nand read.e 0x22000000 KERNEL 0x500000; gpio out 8 1; bootm 0x22000000;nand read.e 0x22000000 RECOVERY 0x500000; bootm 0x22000000
bootdelay=0
bootparts=mtdparts=nand0:0x140000@0x0(UNKNOWN0),0x60000(IPL0),0x60000(IPL1),0x60000(IPL_CUST0),0x60000(IPL_CUST1),0xc0000(UBOOT0),0xc0000(UBOOT1),0x60000(ENV0),0x60000(BOOTL0G0),0x5
00(RECOVERY),0x7100000(UBI)
ethact=sstar_emac
ethaddr=00:73:14:00:00:01
fileaddr=23af42e8
filesize=1fb
gatewayip=172.19.24.1
initrd_block=77
initrd_high=0x21000000
initrd_size=67beed
ipaddr=192.168.1.20
mtddevname=IPL0
mtddevnum=0
mtdids=nand0=nand0
mtdparts=mtdparts=nand0:384k@1280k(IPL0),384k(IPL1),384k(IPL_CUST0),384k(IPL_CUST1),768k(UBOOT0),768k(UBOOT1),384k(ENV0),0x60000(Logo),0x500000(KERNEL),0x500000(RECOVERY),-(UBI)
netmask=255.255.255.0
partition=nand0,0
```

6. 需要注意的是，ubi 文件格式需要额外的空间来存一下 ubi 的信息，所以 ubi 分区不能全部用完，需要预留如下大小的空间用来存放 ubi 信息：

Ubi block num * 2 * page + 4*page

不然在制作母片的时候会报如下 peb 不够的 error：

```
last_partition : nand0,0
env_partition : nand0,0
mtdparts_init 1764 enter
Creating dynamic volume appconfigs of size 5242880
[ubi_create_volume]: search for vacant volume ID
[ubi_create_volume]: not enough PEBs, total=903, avail=40, vol reserved=42
UBI error: ubi_create_volume: cannot create volume 3, error -28
[do_mstar]: command error!
[do_mstar]: command error!
[do_mstar]: command error!
[nandbin:1319]update fail, please check log
[main]: Make spinand bin failed
```

Eg：公版 spinand 128M 的 flash ,ubi 分区总的大小 903 个 block ,需要预留的大小为 >= 903*2*4k + 4*4k

```
Checksum ok!!
```

IDX:	StartBlk:	BlkCnt:	BackupBlkCnt:	PartType:
0:	0, (000000000000)	10, (0X00140000)	0, (000000000000)	0X0020, (CIS)
1:	10, (0X00140000)	1, (0X00020000)	2, (0X00040000)	0X0003, (IPL0)
2:	13, (0X001A0000)	1, (0X00020000)	2, (0X00040000)	0X0003, (IPL1)
3:	16, (0X00200000)	1, (0X00020000)	2, (0X00040000)	0X0001, (IPL_CUST0)
4:	19, (0X00260000)	1, (0X00020000)	2, (0X00040000)	0X0001, (IPL_CUST1)
5:	22, (0X002C0000)	2, (0X00040000)	4, (0X00080000)	0X0006, (UBOOT0)
6:	28, (0X00380000)	2, (0X00040000)	4, (0X00080000)	0X0006, (UBOOT1)
7:	34, (0X00440000)	3, (0X00060000)	0, (000000000000)	0X000D, (ENV)
8:	37, (0X004A0000)	1, (0X00020000)	0, (000000000000)	0XF000, (KEY_CUST)
9:	38, (0X004C0000)	3, (0X00060000)	0, (000000000000)	0X0002, (LOGO)
10:	41, (0X00520000)	40, (0X00500000)	0, (000000000000)	0X0012, (KERNEL)
11:	81, (0X00A20000)	40, (0X00500000)	0, (000000000000)	0X0009, (RECOVERY)
12:	121, (0X00F20000)	903, (0X070E0000)	0, (000000000000)	0X0021, (UBI)

所以在 project\image\configs\i2m\spinand.ubifs.p2.partition.config 中, ubi 分区总的大小不能超过

0X070E0000 - 903*2*4k + 4*4k

```
66 rootfs$(RESOUC) := $(OUTPUTDIR)/rootfs
67 rootfs$(FSTYPE) := ubifs
68 rootfs$(PATSIZE) := 0xA00000
69 rootfs$(BOOTENV) := console=ttyS0,115200 ubi.mtd=UBI,2048 root=ubi:rootfs ro rootfstype=ubifs init=/linuxrc rootwait=1
70
71 miservice$(RESOUC) := $(OUTPUTDIR)/miservice/config
72 miservice$(FSTYPE) := ubifs
73 miservice$(PATSIZE) := 0xA00000
74 miservice$(MOUNTTG) := /config
75 miservice$(MOUNTPT) := ubi0:miservice
76 miservice$(OPTIONS) := rw
77 miservice$(OTABLK) := /dev/ubi0_1
78
79 customer$(RESOUC) := $(OUTPUTDIR)/customer
80 customer$(FSTYPE) := ubifs
81 customer$(PATSIZE) := 0x5000000
82 customer$(MOUNTTG) := /customer
83 customer$(MOUNTPT) := ubi0:customer
84 customer$(OPTIONS) := rw
85 customer$(OTABLK) := /dev/ubi0_2
86
87 appconfigs$(RESOUC) := $(OUTPUTDIR)/appconfigs
88 appconfigs$(FSTYPE) := ubifs
89 appconfigs$(PATSIZE) := 0x500000
90 appconfigs$(MOUNTTG) := /appconfigs
91 appconfigs$(MOUNTPT) := ubi0:appconfigs
92 appconfigs$(OPTIONS) := rw
93 appconfigs$(OTABLK) := /dev/ubi0_3
```

1.2. 执行 SstarMakeBin 生成 nand.bin

需要注意的是在执行打包命令前, 需要将 SpinandPackBinTool\images\scripts\[set_partition.es 下面这行屏蔽掉:

```
1 # <- this is for comment / total file size must be less than 4KB
2 mtdparts del CIS
3 #setenv mtdparts $ (mtdparts) , 0x60000 (LOGO) , 0x500000 (KERNEL) , 0x500000 (RECOVERY) , - (UBI)
4 saveenv
5 nand erase part UBI
6 ubi part UBI
7 ubi create rootfs 0xA00000
8 ubi create miservice 0xA00000
9 ubi create customer 0x5100000
10 ubi create appconfigs 0x300000
11
12 % <- this is end of file symbol
13
```

然后在 linux 执行

```
./SstarMakeBin -n SPINAND.INI
```

这样会生成一个 nand.bin

```
aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTool$ ./SstarMakeBin -n SPINAND.INI
[parse_ini_file:1161]ini file=SPINAND.INI
[iniparser_line][467]: [path]
[iniparser_line][467]: root_directory=./
[iniparser_line][467]: image_directory=images/
[iniparser_line][467]: script_file=auto_update_spinand.txt
[iniparser_line][467]: outpath=./
[iniparser_line][467]:
[iniparser_line][467]: [nand]
[iniparser_line][467]: nandtype=SPINAND
[iniparser_line][467]: nandid=0xC2-0x12
[iniparser_line][467]:
[iniparser_line][467]: [env]
[iniparser_line][467]: env_type=NANDRAW
[iniparser_line][467]: env_part=ENV
[iniparser_line][467]: #env_volume=MPPOOL
[iniparser_line][467]: #env_offset=0
[iniparser_line][467]:
Nand Flash basic information:
  main=2048 bytes(Hex:0x800), spare=64 bytes(Hex:0x40)
  main+spare=2112 bytes(Hex:0x840), page number per block=64
  block size=0x20000(128KB, main), block size=0x21000(main+spare)

[nand_truncate_outfile]: SPINAND type
[nand_truncate_outfile]: truncate file with length 0x1b75000
[nandbin:1355]Nand bin file done!!!

aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTool$
aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTool$ ls
auto_update_spinand.txt  images  leap.def  nand.bin  readme.txt  snchk.def  SPINAND.INI  SstarMakeBin
aaron.feng@sigmastar:~/Code/P2/Beta_Release/mstarbin/release/SpinandPackBinTool$
```

这个 bin 就是烧录程序，然后使用烧录器烧录即可。